

Memo 4: MULTIPLE REACTIONS & SELECTIVITY ANALYSIS
Covering Chapter 4**Due Date: 22/04/2026****OVERVIEW**

Evaluate reactor performance with competing reactions. Using your reactor configuration from Memo 3 (isothermal, fixed geometry), analyze how temperature affects product selectivity when multiple reactions occur simultaneously.

PART 1: PYTHON MULTIPLE REACTION MODEL (75 pts)

Modify your Memo 3 Python model to include multiple reactions:

Required deliverables:**1. Molar Flow Profiles (20 pts)**

- Plot molar flow of each species (A, B, C, D, etc.) vs. catalyst weight
- Use your Memo 3 reactor geometry (D, L, temperature)
- Show all reactants, products, and byproducts on one plot
- Label clearly; include legend

2. Reaction Pathway Diagram (10 pts)

- Diagram showing all reactions in your network
- Example: $A \rightarrow B$ (desired), $A \rightarrow C$ (byproduct), $B \rightarrow D$ (over-reaction)
- Include stoichiometry and reaction rates for each pathway

3. Selectivity vs. Temperature Analysis (35 pts)

- Fix feedstock conversion at constant target value (e.g., $X_A = 0.80$)
- Run reactor model at 3-5 different temperatures
- For each temperature, record:
 - o Conversion X_A (maintain constant)
 - o Selectivity to primary product: $S_B = \text{moles B formed} / \text{moles A reacted}$
 - o Selectivity to byproducts (if applicable)
 - o Yield to primary product
- Create comparison table showing T vs. selectivity
- Plot selectivity vs. temperature (all products on one graph)
- **Discussion:** Which temperature maximizes desired product selectivity? What trade-offs are visible?

4. Assumptions & References (10 pts)

- List all assumptions: isothermal operation, catalyst type, particle size (5 mm unless referenced), reactor config from Memo 3
- References for kinetic parameters (rate constants, activation energies, orders)
- Reference for catalyst particle size (if different from 5 mm)
- All citations in AIChE format

PART 2: POLYMATH VERIFICATION (Optional)

Set up equivalent ODE model in POLYMATH with same multiple reactions and reactor configuration. Compare molar flows and selectivity results with Python model. Results should match within $\pm 1\%$ conversion and $\pm 2\%$ selectivity. Include brief comparison statement.

PART 3: ASPEN PLUS SIMULATION (Optional)

Use RPLUG reactor block with your multiple reaction network and Memo 3 geometry. Report:

- Molar flow profiles vs. reactor length
- Conversion and selectivity results
- Comparison with Python/POLYMATH results
- Brief discussion of any significant differences

PART 4: REPORT QUALITY & COMMUNICATION (25 pts)

Professional presentation:

- All plots clearly labeled with axes, units, legend
- Tables show conversion, selectivity, temperature comparison
- Writing is clear and concise
- Reaction pathway diagram is professional and easy to understand
- All references properly cited

SUBMISSION CHECKLIST

- ☐ Python code (working, all kinetics implemented)
- ☐ Molar flow vs. catalyst weight plot
- ☐ Reaction pathway diagram
- ☐ Selectivity vs. temperature plot
- ☐ Selectivity comparison table
- ☐ Discussion of temperature effects and selectivity trade-offs
- ☐ Complete assumptions and references
- ☐ Optional: POLYMATH model & results
- ☐ Optional: Aspen Plus results & comparison

Format: Single PDF file, `Memo4\_GroupName.pdf`

Late submission: Less than 24 hrs: 50% credit only
More than 24hrs: 0% credit